



MICCAI 2026 · SPOTLIGHT

ConnecToMind2

Inter-Subject fMRI Decoding via Whole-Brain Connectome-Guided Alignment



Gunwoo Baet†, Yeonwoo Kim†, Junha Bae, Mansu Kim*

Gwangju Institute of Science and Technology (GIST), Republic of Korea



Neural Decoding: fMRI to Image

Neural decoding recovers the stimulus a person received from their measured brain activity. Specifically, we reconstruct the visual stimulus that the subject saw from fMRI.



Brain activity

fMRI measured while viewing



Decoding model



Reconstructed image

model output



Prior Decoders Flatten Only the Visual Network

1 Flattened voxel vector

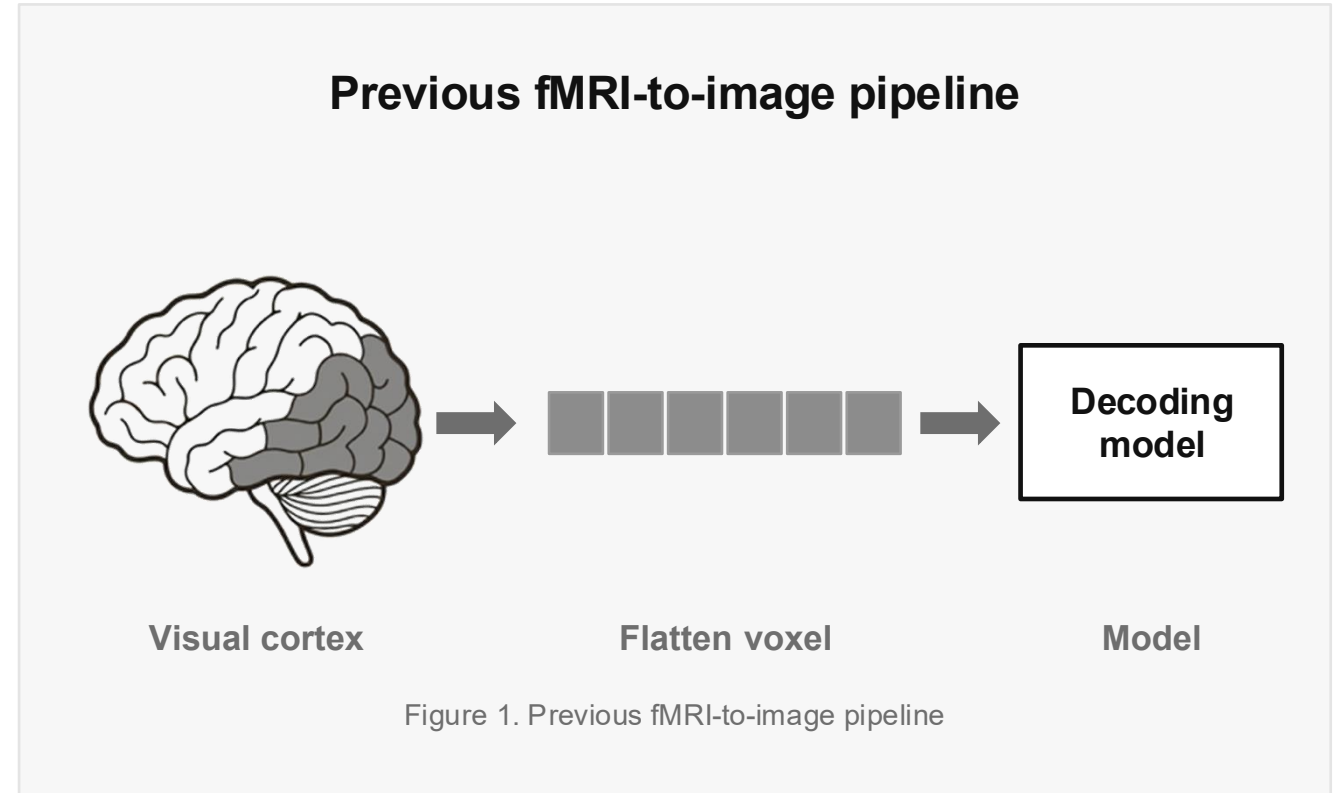
Brain regions already have well-established functional roles, but flattening ignore which region each voxel came from.

2 Visual network only

Visual perception is not solved by the visual network alone; attention and higher-order networks take part in it as well.

3 Subject-specific adaptation

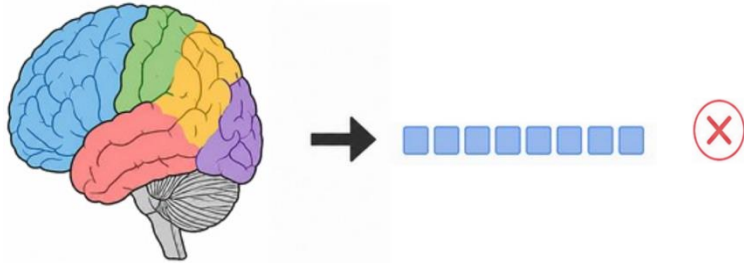
Every subject carries their own noise. A per-subject model absorbs that noise instead of removing it.



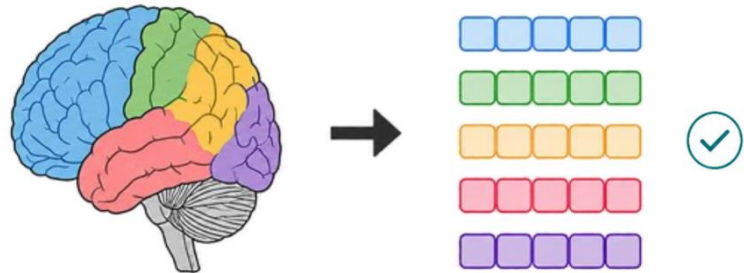


Three Design Choices of ConnectToMind2

1 · Flattened → region-level

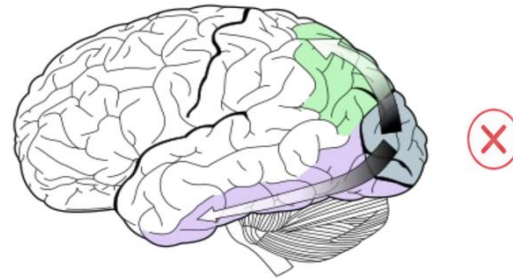


Flattened fMRI Signal



Region-Level Brain Embedding

2 · Visual cortex → whole brain

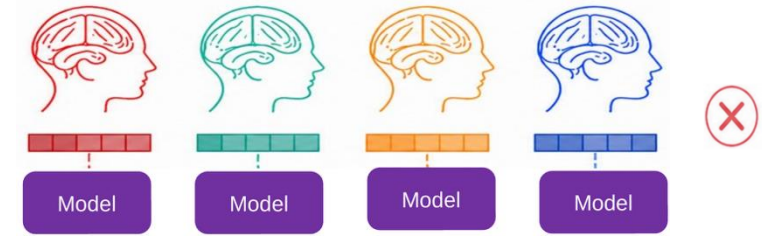


Visual Cortex

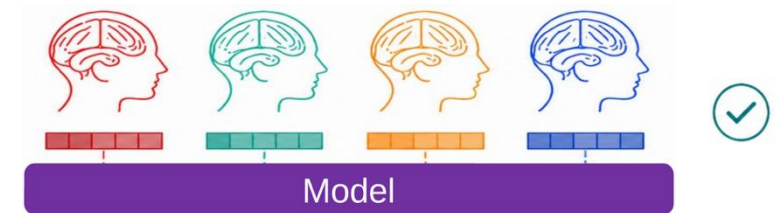


Whole Brain

3 · Per-subject → shared model



Intra-Subject Model



Inter-Subject Model

Figure 2. Top row: what prior work does (X). Bottom row: the design choices of ConnectToMind2 (✓).

ConnecToMind2



- (a) Whole-brain regional embedding gives each ROI its own projection, and a single autoencoder then merges those subspaces into one common space.
- (b) Connectome-guided Q-Former applies cross-attention between the learnable queries and the regional embedding, with structural connectivity added to the attention logits before softmax.

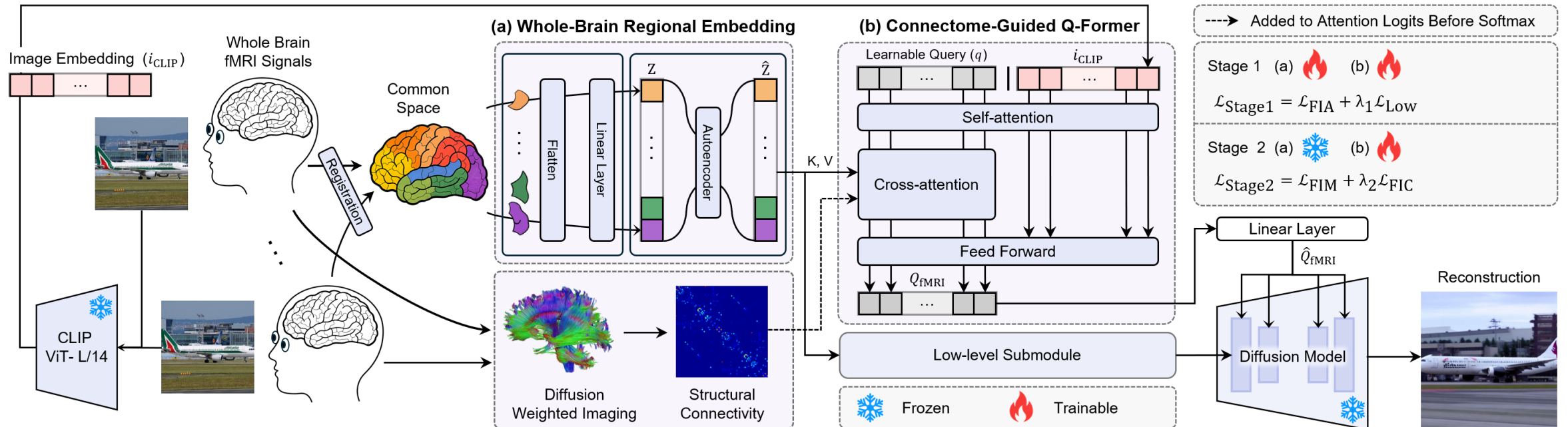


Figure 3. (a) Whole-brain regional embedding, (b) connectome-guided Q-Former with the SC prior, and the frozen diffusion decoder on the right.

Main Experiments



The table and figure show the best low-level and strong high-level performance, proving the impact of our approach.

Setting	Model (Regions)	Low-level				High-level			
		PCorr $\uparrow$	SSIM $\uparrow$	Alex(2) $\uparrow$	Alex(5) $\uparrow$	Incep $\uparrow$	CLIP $\uparrow$	Eff $\downarrow$	SwAV $\downarrow$
Intra	MindEye1 (Visual)	0.167	0.290	88.5%	95.4%	92.0%	91.0%	0.686	0.402
Intra	ConnecToMind (Visual)	0.170	0.297	88.5%	95.9%	93.5%	92.3%	0.666	0.388
Intra	ConnecToMind2 (Whole)	0.278	0.387	87.9%	93.4%	88.1%	89.7%	0.769	0.487
Inter	MindBridge (Visual)	0.091	0.214	75.3%	86.7%	85.2%	80.9%	0.853	0.553
Inter	Umbrae (Visual)	0.054	0.233	70.7%	85.2%	86.7%	88.2%	0.748	0.482
Inter	ConnecToMind2 (Whole)	0.263	0.407	86.5%	92.6%	86.8%	87.1%	0.771	0.487

Table 1. Reconstruction performance on NSD shared-1,000. Intra = per-subject models; Inter = one shared model across subjects.

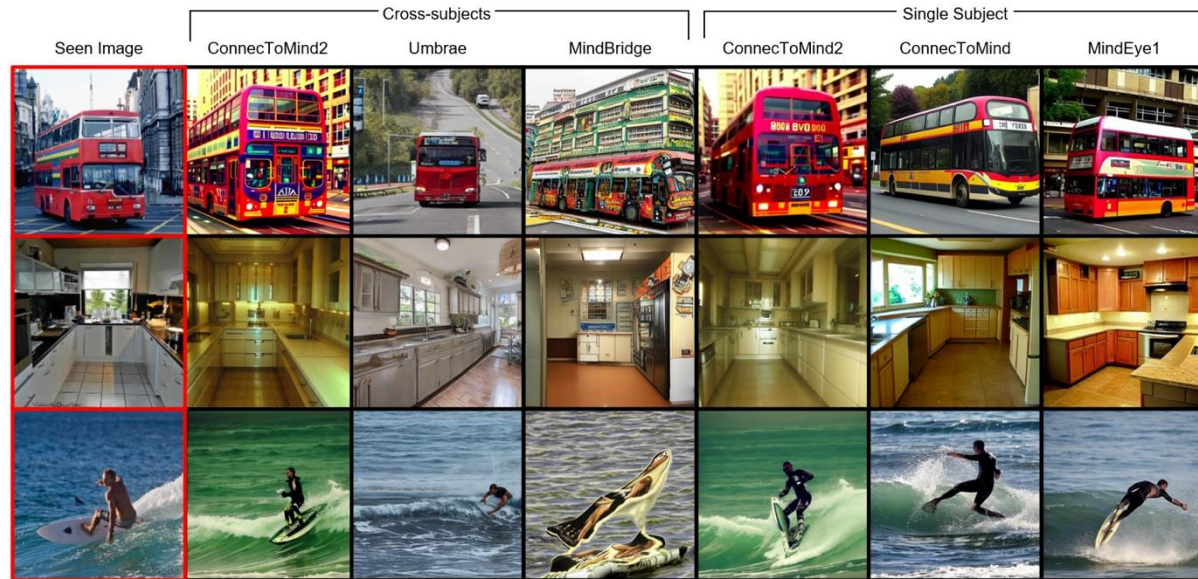


Figure 4. Left column = seen image; next three = cross-subject models; last three = single-subject models.



Ablation Study: Why Regions Must Be Given Region by Region

Removing the regions that handle low-level features degrades only the low-level scores, while removing the higher-order regions degrades only the high-level scores, so regional function carries straight into the output and justifies whole-brain region-level embedding.

Method	Low-level				High-level			
	PCorr $\uparrow$	SSIM $\uparrow$	Alex(2) $\uparrow$	Alex(5) $\uparrow$	Incep $\uparrow$	CLIP $\uparrow$	Eff $\downarrow$	SwAV $\downarrow$
ConnecToMind2	0.263	0.407	86.5%	92.6%	86.8%	87.1%	0.771	0.487
ConnecToMind2 (w/o L)	0.175	0.367	79.5%	88.9%	85.7%	86.2%	0.811	0.517
ConnecToMind2 (w/o H)	0.257	0.418	83.5%	90.0%	84.4%	85.8%	0.799	0.511

Table 2. w/o L = early visual regions removed; w/o H = higher-order regions removed. Shading = worst score per column.

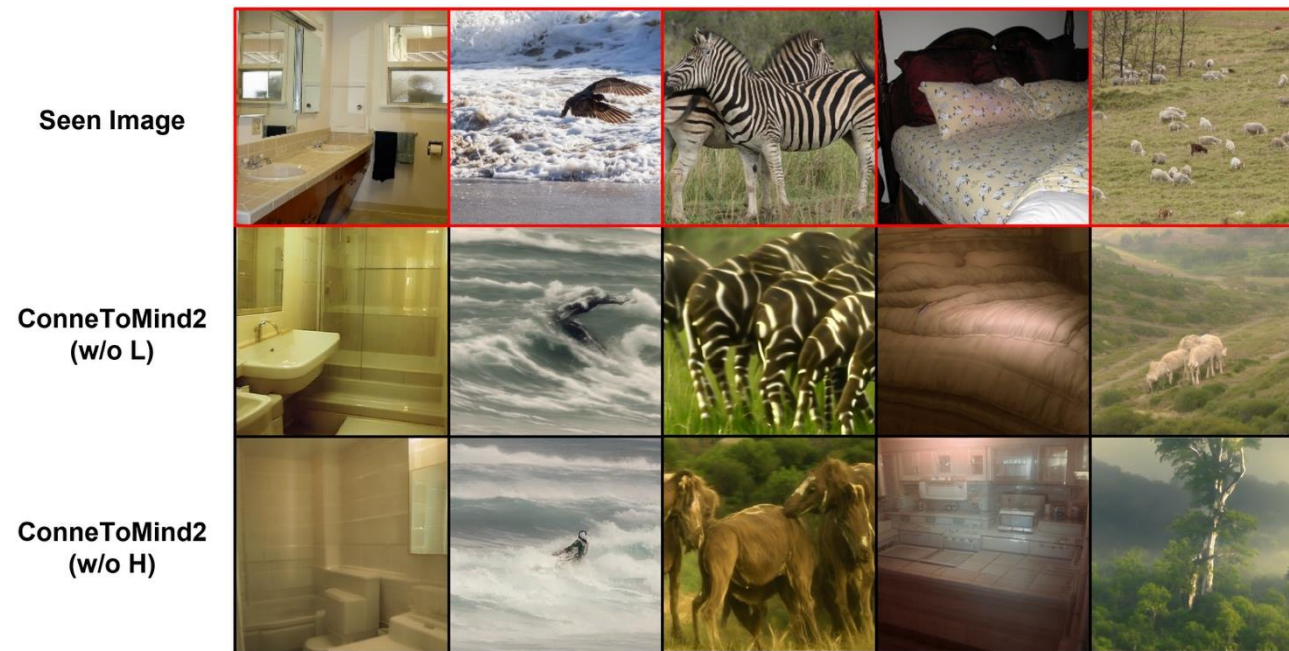


Figure 5. Row 1 = seen image; row 2 = without low-level regions; row 3 = without high-level regions.